(a) Form the joint cavity

$$q(z) = \mathcal{N}(m, V)$$



$$\Lambda_{-j} = \Lambda - \Lambda_j$$

$$\eta_{-j} = \eta - \eta_j$$

$$q_{-j}(z) = \mathcal{N}(m_{-j}, V_{-j})$$

Repeat R rounds ($R = 3$ here)

(b) Predict the sensor reading

Condition on the other sensors



Residual $r_j = \tilde{x}_j - \hat{x}_j$

$$\hat{x}_j = A_j(m_{-j,\,y} - \mu_y) + D_j^{\top} m_{-j,\,h}$$

$$v_j = w_j^{\top} V_{-j} w_j + \psi_j$$

Use r_j, $r_j/\sqrt{v_j}$, and $\log v_j$

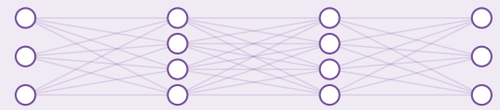
(c) Shared correction network

Support scatter $\rightarrow e_j \in \mathbb{R}^{16}$



DeepSets encoder

Reading, anchor, cavity statistics
Residuals and observed fraction



$$f_\theta : \quad (\delta_j^{\text{off}}, \delta_j^{\text{slope}}, \delta_j^{D}, s_j)$$

(d) Reconstruct the Gaussian site

$$\bar{y} = m_{-j,\,y}$$

$$w_j = (A_j + \delta_j^{\text{slope}},\ D_j + \delta_j^{D})$$

$$g_j(\bar{y}) = A_j(\bar{y} - \mu_y) + \delta_j^{\text{off}}$$

$$o_j = \tilde{x}_j - g_j(\bar{y}) + w_{j,\,y}\bar{y}$$

$$\psi_j = \psi_j^{(0)} \exp\{4\tanh(s_j/4)\} > 0$$

$$\Lambda_j = w_j w_j^{\top}/\psi_j, \quad \eta_j = w_j o_j/\psi_j$$

(e) Fuse and marginalize

Sum the observed replacement sites

$$\Lambda = \Lambda_0 + \sum_{j \in S} \Lambda_j$$

$$\eta = \eta_0 + \sum_{j \in S} \eta_j$$

$$m = \Lambda^{-1}\eta, \quad V = \Lambda^{-1}$$

$$q(y) = \mathcal{N}(m_y, V_{yy})$$

Absent sensors contribute zero

(f) Train with target likelihood

After the final refinement round

$$\mathcal{L} = \mathbb{E}_{\tau,\,\mathcal{D},\,x_S,\,y}[-\log q_\theta(y)]$$



Zero output head recovers the anchor